Question	Answer	Marks	Partial Marks
4(b)	$x = -0.75$ oe	3	**M1** for $15(2x+5)=14(3-x)$ **B1** for $30x+75=42-14x$ or better
4(c)	$3x^2-16x-35[=0]$ or $3y^2-8y-51[=0]$	M3	**M1** for $x^2+2(4-x)^2=67$ or $(4-y)^2+2y^2=67$ seen **B1** for $16-8x+x^2$ or $16-8y+y^2$
	$(3x+5)(x-7)$ [= 0] or $(3y-17)(y+3)$[= 0]	M1	or for correct factors for *their* equation or for correct use of quadratic formula or completing the square for *their* equation
	$x = 7,\ y = -3$ $x=-\frac{5}{3},\ y=5\frac{2}{3}$	B2	**B1** for $x = 7,\ x=-\frac{5}{3}$ or for $y = -3,\ y=5\frac{2}{3}$ or for a correct pair of x and y values
5(a)	$(4x-5)(x+3)+(x+1)(x-3)=342$ or $2x(4x-5)-(3x-6)(x-3)=342$	M2	**M1** for $(4x-5)(x+3)$ or $(x+1)(x-3)$ or for $2x(4x-5)$ or $(3x-6)(x-3)$
	$4x^2+12x-5x-15$ oe and $x^2+x-3x-3$ oe seen OR $8x^2-10x$ and $3x^2-15x+18$ seen	M2	**M1** for each
	$5x^2+5x-18=342$ leading to $x^2+x-72=0$	A1	no errors or omission
5(b)	$(x+9)(x-8)$	M2	**B1** for $(x+a)(x+b)$ where $ab = -72$ or $a + b = 1$ and a, b are integers
	8, –9	B1	
5(c)	86	2	**FT** for $12\times$*their* $x-10$ (x positive) **B1** for any one of 27, 11, 16 seen or for $2x+2x+4x-5+4x-5$ oe or better soi
5(d)	22.2 or 22.16 to 22.17	2	**M1** for tan = $\frac{11}{27}$ or $\frac{\textit{their } x+3}{4\times\textit{their } x-5}$
6(a)(i)	29.5 or 29.50…	4	**M2** for $\frac{11^2+5.3^2-6.9^2}{2\times11\times5.3}$ or **M1** for $6.9^2 = 11^2+5.3^2-2\times11\times5.3\cos x$ **A1** for 0.87[0…] oe

Question	Answer	Marks	Partial Marks
6(a)(ii)	13.4 or 13.38…	**4**	**B1FT** 84 – *their* **(*a*)(i)** **M2** for $\frac{11}{\sin 42} \times \sin \textit{their}\, 54.5$ or **M1** for implicit form
6(b)	2700	**4**	**M2** for $15 \times 2.5 \times 20 \times 60 \times 60$ or **M1** for $15 \times 2.5 \times 20$ **M1** for *their* volume ÷ 1000 If 0 scored, **SC1** for figs 27 with no working
7(a)(i)	$\frac{3}{4}, \frac{1}{4}$ $\frac{2}{5}, \frac{3}{5}$ $\frac{2}{5}, \frac{3}{5}$	**2**	**B1** for one correct pair
7(a)(ii)	$\frac{3}{10}$ oe	**2**	**FT** *their* tree diagram **M1** for $\frac{3}{4} \times \frac{2}{5}$
7(a)(iii)	$\frac{11}{20}$ oe	**3**	**M2** for $\frac{3}{4} \times \frac{3}{5} + \frac{1}{4} \times \frac{2}{5}$ or **M1** for $\frac{3}{4} \times \frac{3}{5}$ or $\frac{1}{4} \times \frac{2}{5}$
7(b)	$\frac{36}{125}$ oe	**3**	**M2** for $\left(\frac{2}{5}\right)^2 \times \frac{3}{5} \times 3$ oe or **M1** for $\left(\frac{2}{5}\right)^2 \times \frac{3}{5}$
7(c)	$\frac{3}{28}$ oe	**2**	**M1** for $\frac{3}{4} \times \frac{1}{7}$
8(a)	12	**2**	**M1** for $150 = \frac{(n-2) \times 180}{n}$ or $\frac{360}{180-150}$ oe
8(b)(i)	45	**2**	**B1** for angles at M or K = 45 or angle at L = 90
8(b)(ii)	85	**2**	**B1** for either angle in alt segment = 58
8(b)(iii)	72	**2**	**B1** for either angle at J or H=108 or angle at F=72
8(c)	$OA = OB = OC = OD$ Radii	**B1**	
	$AB = CD$ chords equidistant from centre are equal	**B1**	
	SSS implies congruent	**B1**	